

Counterexample packet for arXiv:2311.15365

Critical quadratic growth can destroy attainment

Status. `counterexample_likely_valid`. This packet gives a negative answer to the critical- $p = 2$ minimizer-existence question in Noboru Isobe, *A convergence result of a continuous model of deep learning via Łojasiewicz–Simon inequality*, arXiv:2311.15365v2 [1].

Source problem. The source considers, for $\eta \in L^2(0, 1; \mathcal{P}_2(\mathbb{R}^m))$,

$$J(\eta) = \int_{\mathbf{X} \times \mathbf{Y}} \ell \, d\mu_1^\eta + \frac{\lambda}{2} \int_0^1 \int_{\mathbb{R}^m} |\theta|^2 \, d\eta_t(\theta) \, dt, \quad (1)$$

where μ_t^η solves

$$\partial_t \mu_t^\eta + \operatorname{div}_x(\mu_t^\eta v_{\eta_t}) = 0, \quad v_{\eta_t}(x) = \int_{\mathbb{R}^m} v(x, \theta) \, d\eta_t(\theta). \quad (2)$$

Its existence theorem assumes the subcritical bounds, for some $p < 2$,

$$|v(x, \theta)| \leq C|\theta|^p(1 + |x|), \quad (3)$$

$$|v(x_1, \theta) - v(x_2, \theta)| \leq C|\theta|^p|x_1 - x_2|. \quad (4)$$

Remark 7.1 on PDF page 23 says that “the scenario when $p = 2$ in Assumption 1.3 remains unsolved” and asks for an example in which a minimizer does not exist.

REFERENCES

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7. Concluding Remarks

We described the existence and convergence results for a continuous model of DL. We impose [Assumptions 1.3, 1.5, 1.8](#), and [1.9](#) to derive these results. For future studies, we mention the necessity of these assumptions.

Remark 7.1 (For [Assumptions 1.3](#) and [1.5](#)). By virtue of the narrow topology, we show the existence of a minimizer under the weaker assumption ([Assumption 1.3](#)) than [40, Assumption 1.4]. However, the scenario when $p = 2$ in [Assumption 1.3](#) remains unsolved. In addition, the existence of a minimizer in a similar problem in [68, Theorems 2 and 3] is not obvious. It would be considerably intriguing to devise an example where a minimizer does not exist.

Furthermore, relaxing the real analyticity in [Assumption 1.5](#) would be advantageous to ensure convergence in cases such as ReLU.

Figure 1: The open question in Remark 7.1 of arXiv:2311.15365v2, PDF page 23.

Proof intuition. The regularization charges the second moment of the parameter distribution. We choose a vector field whose ratio to this quadratic cost is strictly less than one at every finite parameter but tends to one at infinity. A mass of order M^{-2} placed at $\theta = M$ then has a fixed quadratic cost and a nonvanishing effect on the terminal state. The limiting effect-to-cost ratio is never achieved by a finite parameter distribution, so the objective has a minimizing sequence but no minimizer.

Theorem 1 (Explicit nonattainment at $p = 2$). *In the source problem, take*

$$d = m = 1, \quad \mathbf{X} = \mathbb{R}, \quad \mathbf{Y} = \{0\}, \quad \mu_0 = \delta_{(0,0)}, \quad \lambda = 2,$$

and define

$$v(x, \theta) = \frac{\theta^4}{1 + \theta^2}, \quad \ell(x, 0) = (x - 1)^2. \quad (5)$$

Then all assumptions in the source's minimizer-existence problem hold with the critical exponent $p = 2$, but

$$\inf_{\eta \in L^2(0,1;\mathcal{P}_2(\mathbb{R}))} J(\eta) = \frac{3}{4}$$

and the infimum is not attained.

Proof. First check the hypotheses. The set $\mathbf{Y}$ and the support of μ_0 are compact, $\lambda > 0$, and ℓ is continuous, nonnegative, and

$$\ell(x, 0) = (x - 1)^2 \leq 2 + 2x^2.$$

The vector field is continuous and independent of x . Moreover,

$$0 \leq v(x, \theta) = \theta^2 \frac{\theta^2}{1 + \theta^2} \leq \theta^2,$$

whereas $v(x_1, \theta) - v(x_2, \theta) = 0$. Thus both bounds in (3) hold with $p = 2$ and $C = 1$.

Fix an arbitrary admissible η , and put

$$R = \int_0^1 \int_{\mathbb{R}} \theta^2 d\eta_t(\theta) dt, \quad A = \int_0^1 \int_{\mathbb{R}} \frac{\theta^4}{1 + \theta^2} d\eta_t(\theta) dt. \quad (6)$$

Admissibility gives $R < \infty$, and $0 \leq A \leq R$. Since the averaged vector field in (2) is independent of x , its flow is translation by

$$x_t = \int_0^t \int_{\mathbb{R}} \frac{\theta^4}{1 + \theta^2} d\eta_s(\theta) ds.$$

Starting from $\mu_0 = \delta_{(0,0)}$, this gives $\mu_1^\eta = \delta_{(A,0)}$. Hence, because $\lambda/2 = 1$,

$$J(\eta) = (A - 1)^2 + R. \quad (7)$$

There is a strict pointwise efficiency loss at every nonzero finite parameter:

$$R - A = \int_0^1 \int_{\mathbb{R}} \frac{\theta^2}{1 + \theta^2} d\eta_t(\theta) dt. \quad (8)$$

If $R > 0$, then η_t is not concentrated at zero for a set of times of positive measure, and (8) is strictly positive. Thus $A < R$. If $R = 0$, then $A = 0$ and $J(\eta) = 1$. In the remaining case,

$$\begin{aligned} J(\eta) &> (A - 1)^2 + A \\ &= \left(A - \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4}. \end{aligned}$$

Consequently every admissible η satisfies $J(\eta) > 3/4$.

It remains to show sharpness. Put $r = 1/2$, and for every $M \geq 1$ take the constant-in-time curve

$$\eta_t^M = \left(1 - \frac{r}{M^2}\right) \delta_0 + \frac{r}{M^2} \delta_M, \quad 0 \leq t \leq 1. \quad (9)$$

It belongs to $L^2(0, 1; \mathcal{P}_2(\mathbb{R}))$, and its quantities in (6) are

$$R_M = r = \frac{1}{2}, \quad A_M = \frac{r}{M^2} \frac{M^4}{1 + M^2} = r \frac{M^2}{1 + M^2} \longrightarrow r = \frac{1}{2}.$$

Therefore

$$J(\eta^M) = (A_M - 1)^2 + \frac{1}{2} \longrightarrow \left(\frac{1}{2} - 1\right)^2 + \frac{1}{2} = \frac{3}{4}.$$

The lower bound and this sequence prove that the infimum is $3/4$; its strictness for every admissible η proves nonattainment. $\square$

Verification notes. The proof uses no external theorem beyond the source’s formulation of the continuity-equation model. The flow calculation can also be checked directly in distributions because the time-dependent velocity is spatially constant and integrable. The strict step is exact: the integrand in (8) vanishes only at $\theta = 0$. The included script evaluates the exact formulas for sample M ; it is a smoke test, not part of the proof.

Why the critical exponent matters. The measures in (9) converge narrowly to δ_0 , but their second moments stay equal to $1/2$. At every finite M , velocity is strictly smaller than quadratic cost; the missing efficiency appears only at infinity. For $p < 2$, a bounded second moment controls this lower-order effect uniformly at infinity. At $p = 2$, the effect and the regularizer have the same homogeneity, permitting precisely this escape-of-mass mechanism.

Novelty check and limitations. On 2026-07-21, the run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:2311.15365, the exact title, “continuous DNN,” “ $p = 2$,” “quadratic growth,” and “minimizer does not exist.” Bounded primary-source searches on arXiv used the exact id together with the same phrases and close “continuous model of deep learning” variants. They found the source paper, currently v2, but no later paper or revision resolving this question. This supports candidate novelty but is not an exhaustive claim about all published literature. The theorem addresses minimizer existence at the critical growth exponent; it does not address the source’s separate questions about analyticity, curvature, or convergence of gradient flows.

Human-review recommendation. Review as a full negative answer to the critical- $p = 2$ existence question and as the explicit nonattainment example requested in Remark 7.1. The two points most worth auditing are the translation formula for μ_1^η and the implication $R > 0 \Rightarrow R - A > 0$.

References

- [1] N. Isobe, *A convergence result of a continuous model of deep learning via Łojasiewicz–Simon inequality*, arXiv:2311.15365v2, 2024.